

GROUP 5

HISTOLOGY TO GENE EXPRESSION

Siyu (Bobby) Wang, Mobai (James) Jiang,
Junewoo Koo, Jiayin (Moria) Li



QUICK RECAP

GOAL

Predict the spatial gene-expression map from stained tissue photos.

Like predicting a city's temperature from a satellite photo.

DATASET

HEST

Human tissue and spatial expression data

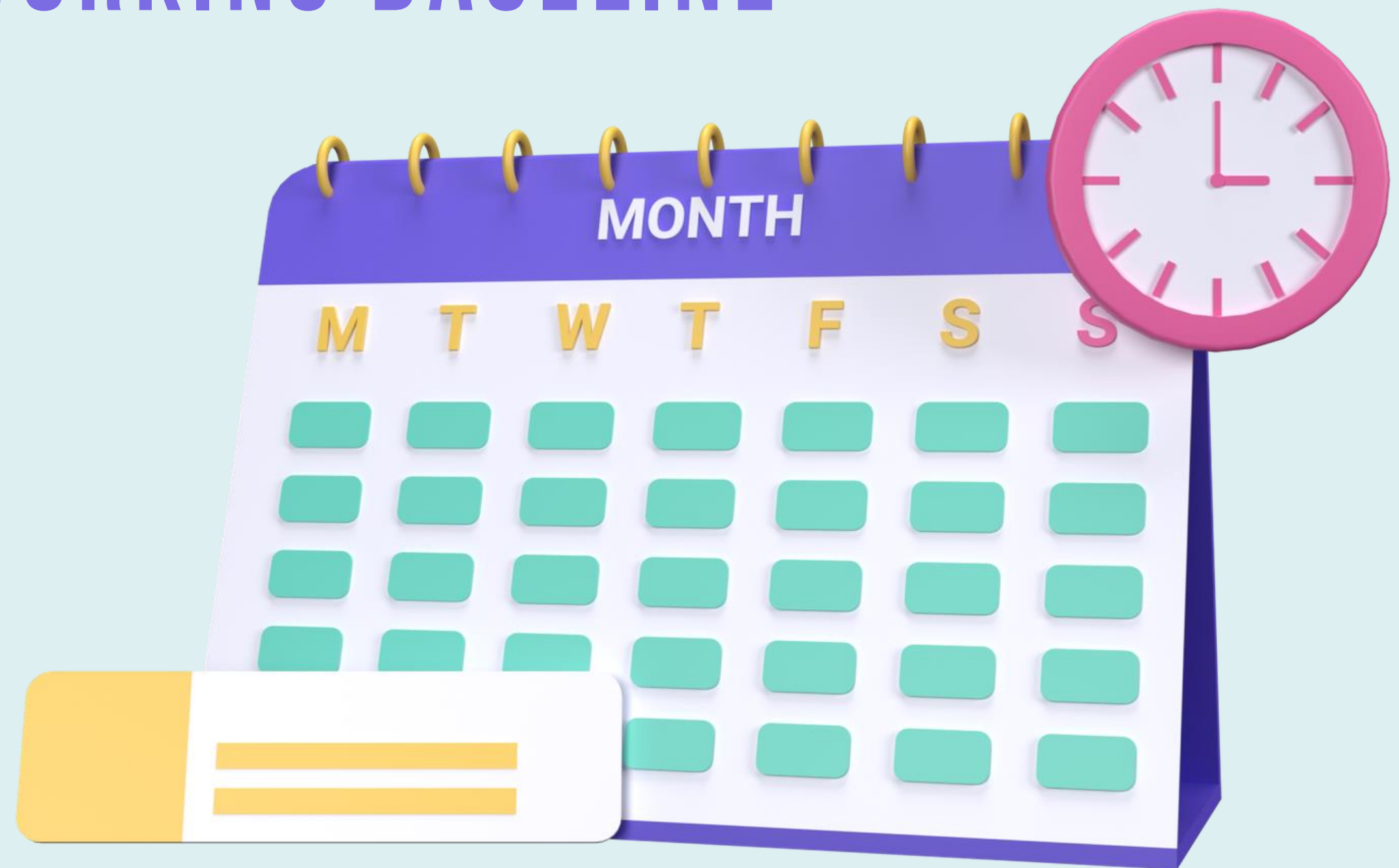
MAKE A WORKING BASELINE

DATA SUBSETS

GENBIO-VIT EMBEDDING

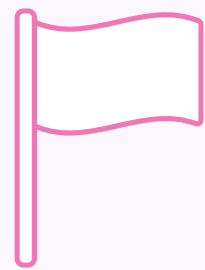
PCA

METRICS & RESULTS



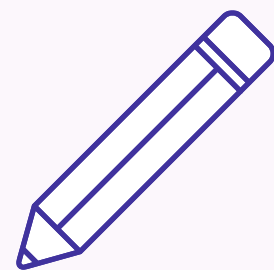
DATASET OVERVIEW

SPECIES



Pick human samples only.

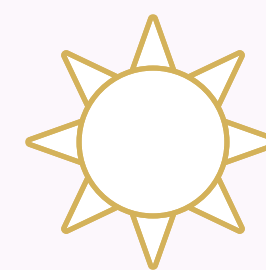
TISSUE



Breast tissues only.
It contains the most patients and slides.

- 31 patients
- 108 samples
- 3 studies

TECHNOLOGY



- Spatial Transcriptomics
- Visium
- Xenium [ZEE-nee-uhm]

GENE TARGETS



- 500+ genes in original dataset
- keep the top 50 ranked by variation (among samples)

SPLITTING THE DATA

UNSEEN

Make the model generalize on patients absent from training

LOW COST

Utilize reasonable amount of data while keeping computation cost reasonable

REPRODUCIBLE

Scripts created to reduce numbers

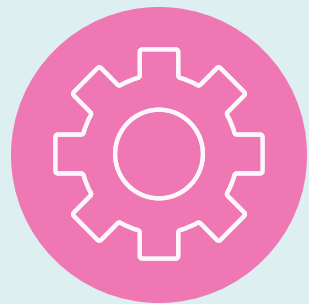


THREE WAYS TO SPLIT



SPOT BASED

Nearby spots can share tissue patterns. Highest leakage when location is provided.



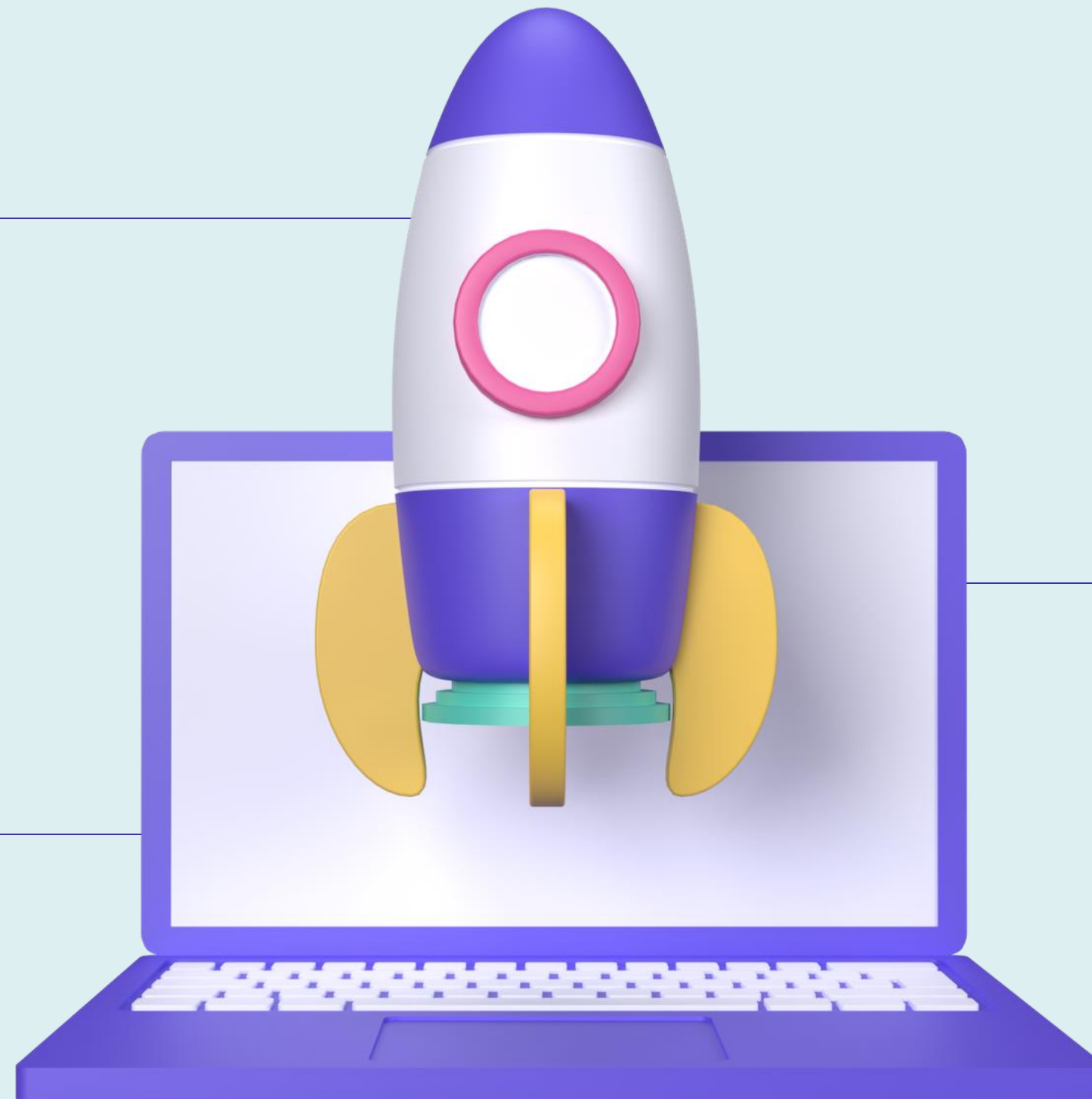
SLIDE BASED

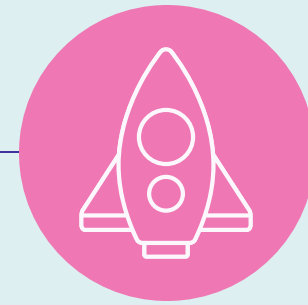
Patients may overlap: more leakage than patient split.



PATIENT BASED

Hold out all samples from a patient.
Least leakage.





SLICE



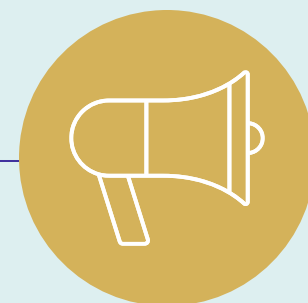
Takes a patch (224x224) and treats it as a grid of 16x16 subunits like visual tokens.



ATTN



Transformer attention combines information across subunits.

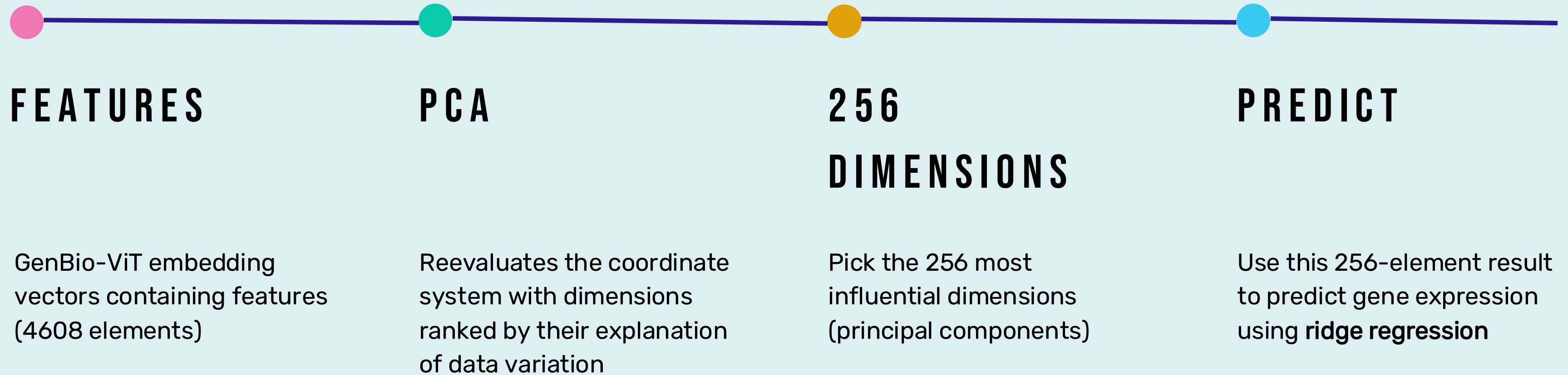


PUBLIC



Foundation model pretrained on massive public data; high performance on HEST benchmark

PCA: COMPRESS THE EMBEDDINGS



METRICS

REAL EXPRESSION VALUES

Observed UMI counts

PREDICTED EXPRESSION VALUES

Predicted UMI counts

UMI=unique molecular identifier • Apply $\log_{1p}(x) = \ln(1 + x)$ to express them on a log scale

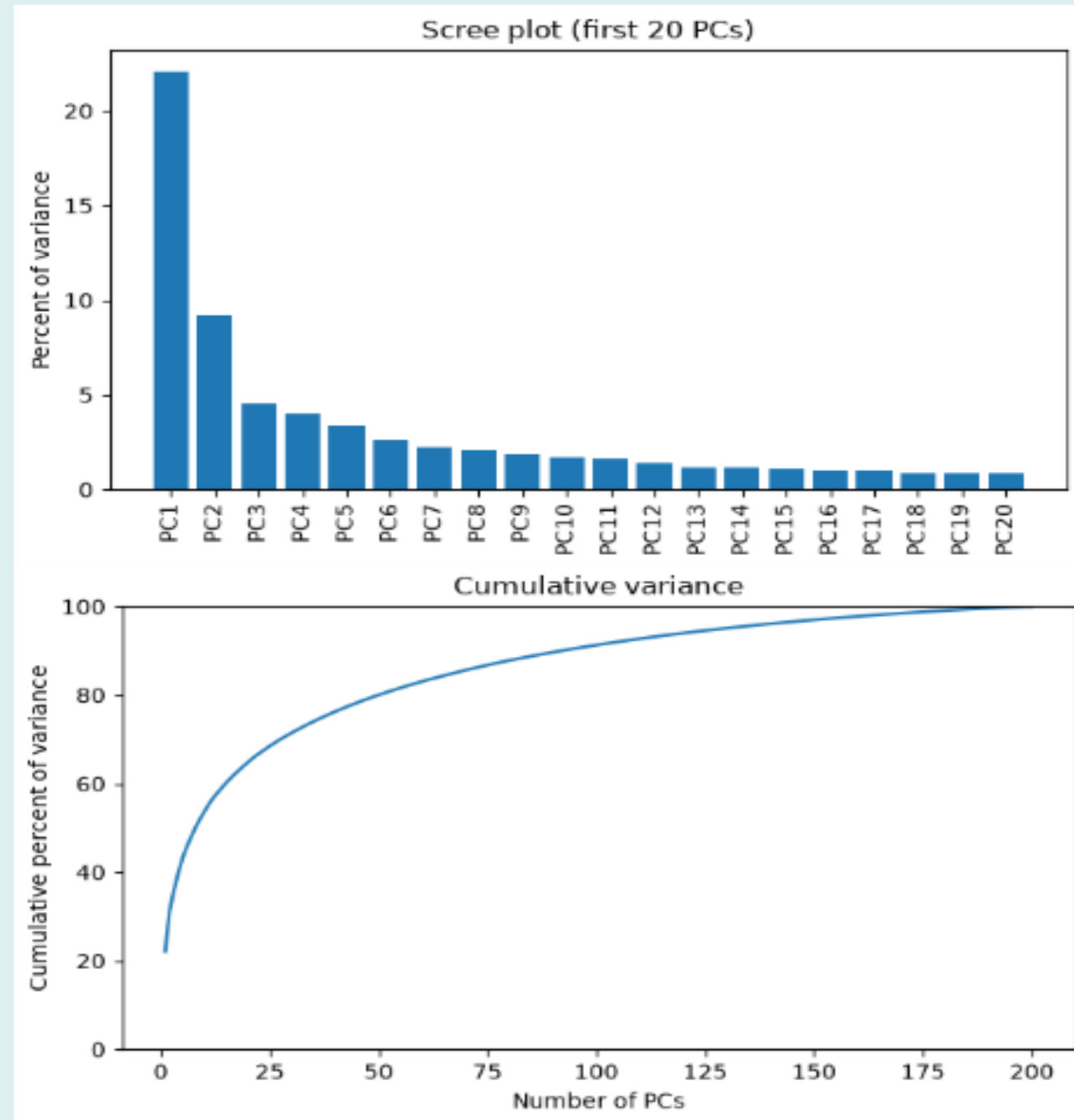
N-RMSE

Normalized root mean squared error across samples. Measures absolute prediction error on $\log_{1p}$ expression values.

PCC

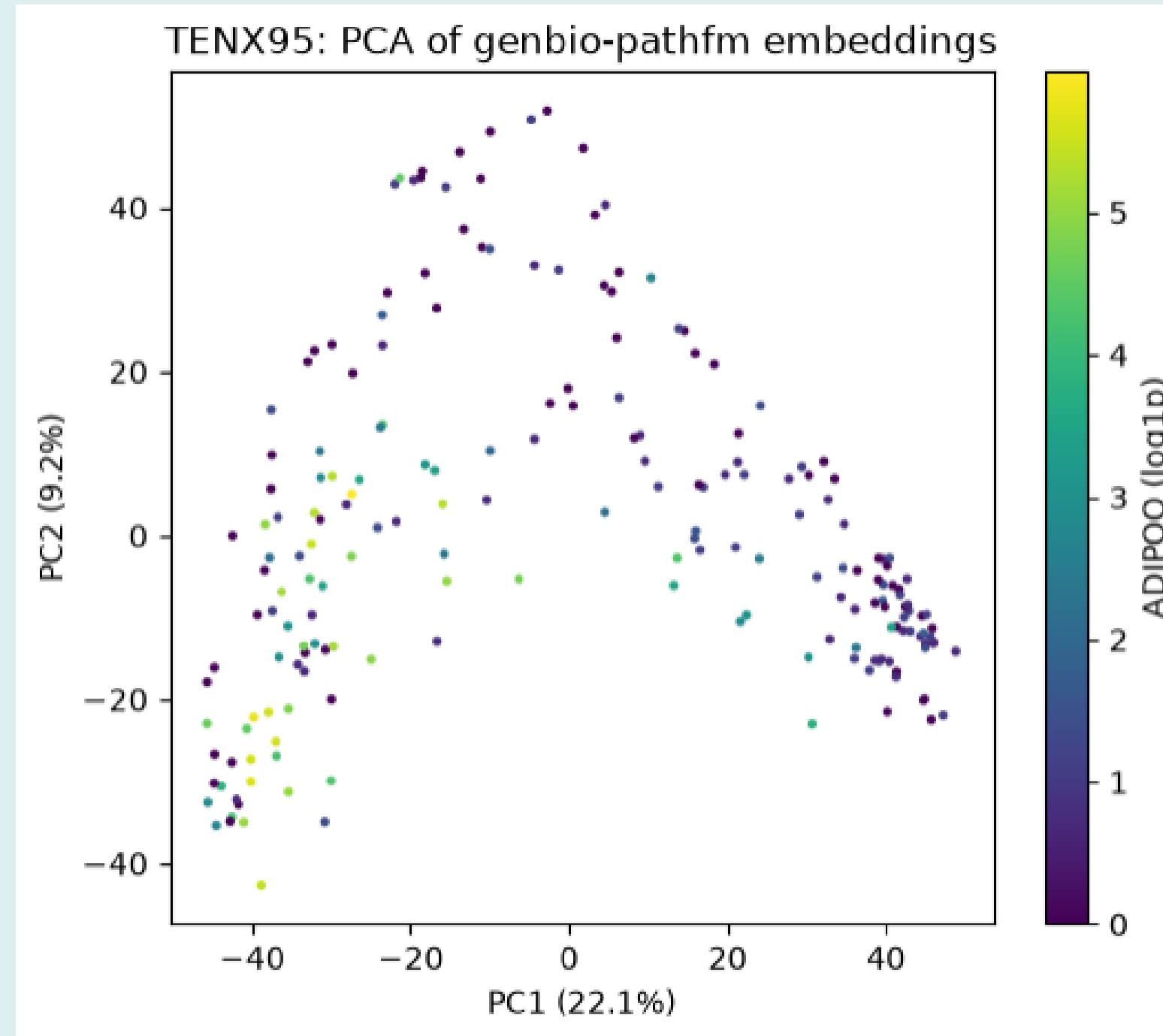
Correlation between real and predicted values. Highlights shared trends, not absolute magnitude.

RESULTS & DISCUSSION



PCA

PCA for the embedding of ID TENX95, trained with 200 randomly picked spots



PCA VISUALIZATION

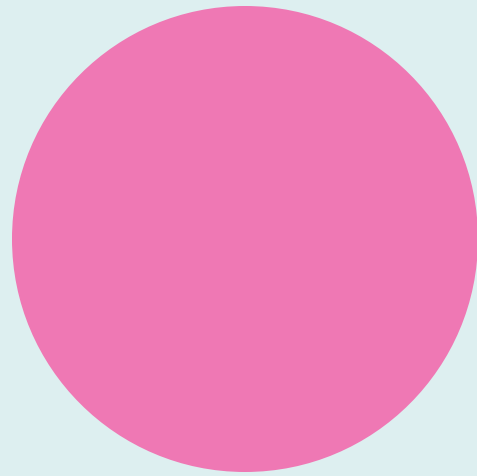
- PCA of ID TENX95 visualized, where color represents log1p of ADIPOQ gene expression (common in fat cells)
- Some separation observed but also group distribution chaotic

PCC

0.42

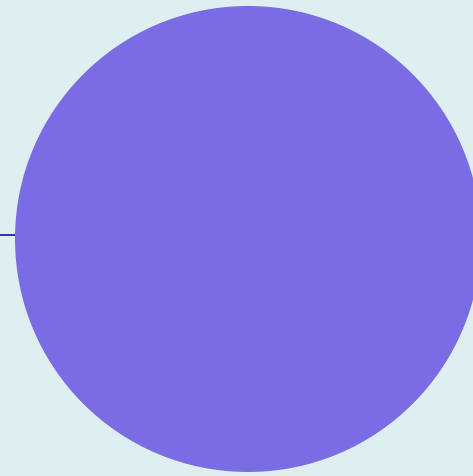
- For the the 200 spots of TENX95

DATA PROCESSING ARCHITECTURE



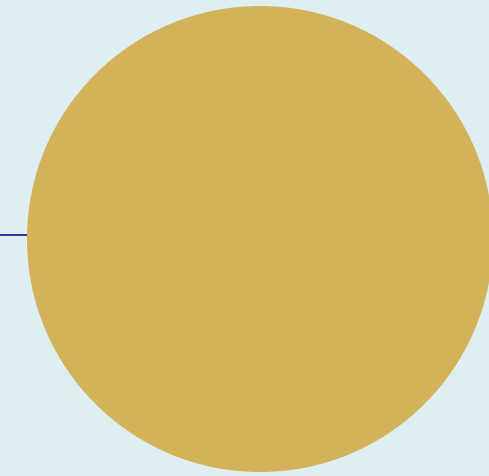
QUERY BATCH

Make one time query for a set batch of data



ON BATCH PROCESSING

Run foundational inference on set batch and delete all unnecessary files



TENSORFICATION

Organize data in .pt format.

```
[ ] import hest_preprocessing
importlib.reload(hest_preprocessing)

from hest_preprocessing import (
    random_split_data,
    export_embeddings,
    export_gene_tensor,
)

Mounted at /content/drive
['hest_preprocessing.py']

[ ] from hest_preprocessing import random_split_data

train_paths, test_paths = random_split_data(
    data_folder="/content/drive/MyDrive/HEST_clean",
    train_percentage=80, # 80% training, 20% testing
    seed=42, # None for a different random split each run
)

Training: 5472 patches; testing: 1368 patches.
Skipped 1 incomplete patches.
Shuffled lists saved in: /content/drive/MyDrive/HEST_clean/splits/split_5d3a5c1db610

[ ] import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/HEST_clean/splits/split_5d3a5c1db610/train.csv", nrows=50)
df.head()
df.to_csv('/content/drive/MyDrive/HEST_clean/splits/split_5d3a5c1db610/trainT.csv')

[ ] embeddings = export_embeddings(
    "/content/drive/MyDrive/HEST_clean/splits/split_5d3a5c1db610/trainT.csv",
    output_path="/content/drive/MyDrive/train_embeddings.pt",
    batch_size=128,
)

Loading weights: 100%  607/607 [00:00<00:00, 2753.31it/s]
Saved [500, 4608] tensor: /content/drive/MyDrive/train_embeddings.pt

[ ] genes = export_gene_tensor(
    "/content/drive/MyDrive/HEST_clean/splits/split_5d3a5c1db610/trainT.csv",
    output_path="/content/drive/MyDrive/train_genes.pt",
    embedding_path="/content/drive/MyDrive/train_embeddings.pt",
)

Saved [100, 18085, 2] tensor: /content/drive/MyDrive/train_genes.pt

[ ] from google.colab import drive
drive.mount('/content/drive')
```

And excerpt of command usage
that does a randomized split and
assignment along with providing
the training data tensor over a
batch size of 500

THANKS FOR WATCHING

Raise questions!

